Recognition and Response to Postpartum Preeclampsia in the Emergency Department (PPED) Initiative

Preliminary Summary

August 2024

Acknowledgments

Members of the PPED workgroup provided expertise and direction throughout the PPED initiative, including James Hill, MD; Gloria Delgado, MSN, RNC-OB; Catherine Eppes, MD, MPH; Patrick Ramsey, MD, MSPH; Susan Dimitrijevic, BSN RN; Joanne Delk, MS; Sue Butts-Dion, MS; and Divya Patel, PhD along with others who provided their expertise. TCHMB acknowledges the contributions of the hospitals that participated in the PPED initiative.

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Background

Preeclampsia is a pregnancy-specific condition that can affect multiple organ systems and is a leading cause of maternal and neonatal mortality worldwide. It is characterized by elevated blood pressure, typically defined as a systolic blood pressure of 140 mm Hg or higher and/or a diastolic blood pressure of 90 mm Hg or higher. Early recognition of preeclampsia symptoms is critical, as it can be lifesaving.[1]

Postpartum preeclampsia is the development or worsening of preeclampsia in the postpartum period, mostly developing within the first 48 hours following birth and up to six weeks later. The 2020 Texas Maternal Mortality and Morbidity Review Committee (MMMRC) and Texas Department of State Health Services (DSHS) joint biennial legislative report identified preeclampsia or eclampsia as the leading cause of death for 11% of reviewed pregnancy-related deaths. Notably, half of these deaths occurred within 42 days of delivery (postpartum period).[2] The report also highlighted that early diagnosis and treatment of preeclampsia and eclampsia could have potentially prevented these deaths.

Further, the 2022 MMMRC report underscored that "emergency health providers' knowledge about maternal physiology and health management, as well as communication and coordination with obstetric and women's health professionals, can be a critical factor in maternal health outcomes."[2] The report also highlighted significant disparities in maternal morbidity and mortality by race and ethnicity, particularly among non-Hispanic Black and Hispanic women. Non-Hispanic Black patients, for instance, have a higher rate of hypertensive disorder-related delivery hospitalizations compared to women of other races and ethnic groups.[3] Additionally, low rates of postpartum visit attendance in Texas limit prevention opportunities and exacerbate disparities. In 2019, 14.8% of Black and 15.1% of Hispanic women had no postpartum checkup, compared to only 5.7% of White women (Texas PRAMS data).[4]

Project Overview and Aims

In response to the findings from the 2020 and 2022 Texas MMMRC reports, the Texas Collaborative for Healthy Mothers and Babies (TCHMB) launched the Recognition and Response to Postpartum Preeclampsia in the Emergency Department (PPED) quality improvement (QI) initiative. This initiative is particularly notable as it represents the first collaboration for TCHMB that includes a unit outside of an obstetric or neonatal unit. By increasing screening for postpartum patients and fostering collaboration between the emergency department (ED) and obstetric (OB) departments, the PPED initiative seeks to decrease the risk for maternal morbidity and mortality from postpartum preeclampsia and reduce disparities among patients presenting to the ED with postpartum preeclampsia.

The two primary aims of the PPED project included:

1. **Improvement of Screening and Diagnosis**: Ensure all patients screened for postpartum status will be screened for elevated blood pressure SBP ≥ 140 mm Hg and/or DBP ≥ 90 mm Hg.
2. **Enhanced Communication and Collaboration**: All joint ED and OB units will perform case reviews for all cases of postpartum patients presenting with elevated blood pressure SBP ≥ 140 and/or DBP ≥ 90 through the emergency department.

Aligned with these aims, and in response to the MMMRC report recommendations, the PPED initiative outlined five main goals for improvement:

1. Identifying postpartum patients presenting to the ED with severe hypertension or preeclampsia.
2. Treating patients identified with severe hypertension in a timely fashion.
3. Improving communication and coordination of care between emergency medicine and obstetric health care teams.
4. Reducing complications from postpartum preeclampsia that lead to maternal morbidity and mortality.
5. Reducing racial disparities in health outcomes for Black and/or Hispanic postpartum patients with severe hypertension or preeclampsia.

These goals are designed to ensure that the initiative addresses the immediate clinical needs of postpartum patients and contributes to the broader objective of reducing disparities in maternal health outcomes across Texas. Full details can be found on the website: https://www.tchmb.org/pped.

Methodology

This project was implemented through TCHMB's partnership with the Regional Advisory Council (RAC) Perinatal Care Region (PCR) Alliance and coordinated with the TexasAIM implementation of the Hypertension Bundle. Hospital recruitment for the PPED project began in August 2022, with the project kickoff webinar taking place in September 2022.

Hospitals across Texas were recruited to participate in the PPED initiative, which aims to reduce postpartum preeclampsia-related maternal morbidity and mortality. To ensure comprehensive statewide coverage, one key goal of TCHMB was to enroll at least one hospital from each of Texas' 22 RACs in the project.

TCHMB leveraged its alliance with the RAC-PCR to facilitate hospital recruitment. RAC-PCR representatives, who serve on the TCHMB OB committee, played a crucial role in disseminating information about the project to their respective RACs, thereby assisting with recruitment efforts across the state.

Additionally, members of the TCHMB PPED Workgroup presented the project in person to hospitals during the TexasAIM Plus Severe Hypertension in Pregnancy (HTN) Learning Sessions held in February and March 2023, further expanding awareness and encouraging participation in the initiative.

The rollout of the PPED project included two cohorts: a Reinforcement Cohort and a Technical Assistance Cohort. The Reinforcement Cohort consisted of hospitals that actively participated in the project, using quality improvement methods and tools to test and implement best practices. These hospitals were deeply engaged, participating in monthly collaborative calls, assisting TCHMB during project workshops at TexasAIM Learning Sessions, and developing and sharing tools and materials specific to the recognition and response to postpartum preeclampsia in the ED. To support these efforts, TCHMB's QI nurse conducted regular check-ins with Reinforcement Cohort hospitals to discuss progress and provide technical assistance as needed. Additionally, these hospitals contributed de-identified patient-level data through the TCHMB REDCap portal. The Technical Assistance Cohort included all other TexasAIM participating hospitals, which received support through the TexasAIM Learning Collaborative. This cohort will also benefit from a toolkit to be developed at the conclusion of the PPED project, which will compile best practices and resources to further assist hospitals in improving maternal and postpartum care.

To meet the goals of the project, Reinforcement Cohort hospitals provided education to ED clinicians, improved processes related to the recognition, diagnosis, and treatment of hypertensive disorders, facilitated better communication and collaboration between emergency and maternal departments, and developed and disseminated clinical management tools for ED clinicians. These efforts were complemented by training staff on accurately documenting self-identified race, ethnicity, and primary or preferred language. Furthermore, Reinforcement hospital members were required to complete the March of Dimes "Awareness to Action: Dismantling Bias in Maternal and Infant Healthcare" training, underscoring the project's commitment to addressing disparities in maternal health.

Key activities of the PPED project included a kickoff webinar, monthly data collection and feedback through PPED data reports, monthly reinforcement calls, case reviews, dissemination of resources on the TCHMB website, and individualized technical support. The figure below illustrates the timeline for these key activities throughout the project.

Figure 1: Timeline of key PPED activities

Enrollment, Participation, and Data Submission

A total of 58 hospitals across Texas completed the enrollment form, 39 of which signed the Data Use Agreement (DUA), and 37 completed the DSUA and PPED Intake Assessment form, making them fully enrolled in the project. The 37 fully enrolled hospitals represented 15 of the 22 Texas Trauma RACs (see Figure 3 below). Of the fully enrolled hospitals, 28 submitted at least one data cycle.

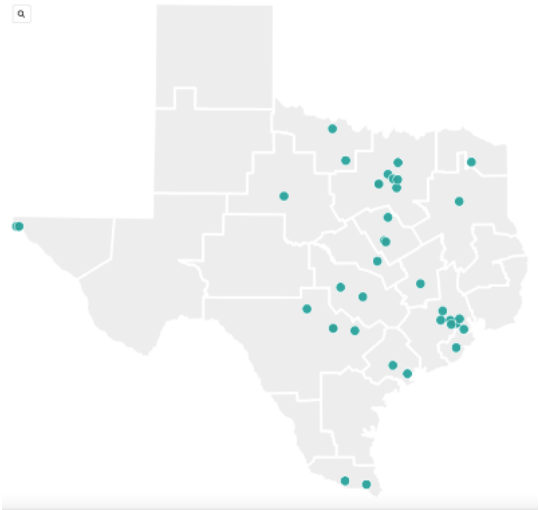

Figure 2: Dots represent the location of hospitals enrolled in the PPED project; white lines represent the outline of each RAC.

Hospitals enrolled in the PPED project were asked to submit monthly data to TCHMB between April 2023 and May 2024. 601 patient records were submitted during the 12 months of data collection. Data metrics included sociodemographic measures such as race, ethnicity, age, language, and insurance; outcome and process measures such as previous diagnosis of hypertension or preeclampsia, previous discharge home on medication for elevated blood pressure/preeclampsia, screening for postpartum status upon arrival to the ED; blood pressure measurement(s) in the ED; symptoms of postpartum preeclampsia; diagnosis and treatment in the ED; and structure measures including the date and time of OB consult for elevated blood pressure and symptoms of postpartum preeclampsia in the ED, patient disposition from the ED, and joint ED and OB case reviews.

To establish a baseline for quality improvement, hospitals were initially asked to submit data consisting of a six-month sampling (September 2022 through February 2023) of patient records from the ED involving cases of postpartum preeclampsia. This process required the PPED data point person at each hospital to meticulously review ED patient records to identify those admitted with signs and symptoms of postpartum preeclampsia.

Data submission was conducted using REDCap, a secure web application for building and managing online surveys and databases, to ensure Health Insurance Portability and Accountability Act (HIPAA) compliance. Hospitals were required to provide both aggregate and de-identified patient-level data monthly through REDCap.

For aggregate data, hospitals identified specific aims, including SMART goals, they wished to achieve through their participation in the project. They also entered information on their Plan-Do-Study-Act (PDSA) focus for the month and any relevant patient information.

De-identified patient-level data was submitted for the subset of postpartum patients presenting to the ED with elevated blood pressure and/or a diagnosis of preeclampsia. These data were used to track patient outcomes and processes across the participating hospitals.

Key Findings and Lessons Learned

Two main improvement opportunities encountered in the project were increasing routine screening for postpartum status in the ED and integrating postpartum status screening into the EHR systems of participating hospitals. While it is common practice to ask women of childbearing age about their current pregnancy status, the absence of follow-up questions regarding recent pregnancies (within the last six weeks) made it challenging to identify postpartum patients at risk for complications like preeclampsia.

To address this gap, participating hospitals collaborated with their information technology (IT) departments to add postpartum status screening to their electronic health records (EHR) systems. However, due to the complexity of updating EHRs, implementation was slower than anticipated, with some hospitals still awaiting the integration of this feature. Consequently, hospitals adopted temporary solutions, such as using paper forms or instructing ED staff to manually review patient information based on age, pregnancy status, and related diagnoses. Although these alternatives were less efficient than pulling data directly from an updated EHR, they enabled hospitals to track the number of

postpartum women visiting the ED—a crucial metric for the project. The lesson learned from these findings was the importance of streamlining health IT processes to enhance patient care and the need for flexibility and creativity in problem-solving when faced with delays or barriers in system updates.

Another key opportunity for improvement was fostering stronger collaboration between obstetric (OB) and emergency department (ED) teams. The project placed a strong emphasis on enhancing communication and coordination between these departments to improve the care of postpartum women. Many hospitals made strides in this area, with one particularly successful hospital sharing its strategies for building a robust partnership between OB and ED teams during a reinforcement call. Their presentation, included in Appendix A, provided valuable insights and best practices, demonstrating that proactive efforts can bridge departmental gaps and lead to more effective, coordinated care.

Additionally, the project underscored the potential for increased patient involvement in improvement initiatives. Partnering with organizations such as the Institute for Patient and Family Centered Care and MoMMA's Voices, participating hospitals were provided with tools and resources to engage patients and families in their improvement efforts. While some hospitals faced challenges, such as securing administrative support or identifying patient partners, others successfully integrated patient perspectives, which added meaningful insights to their QI work. This highlighted the critical role of patient collaboration in addressing important issues like patient safety, healthcare delivery, and reducing racial disparities.

As the project progressed, ED staff gained a deeper understanding of the unique physiological considerations for pregnant and postpartum women, particularly the lower thresholds for elevated and severe range blood pressure that indicate preeclampsia. This new knowledge reinforced the importance of tailored education and ongoing training to ensure ED personnel remain vigilant in identifying and treating preeclampsia promptly. The focus on continuous education fostered an environment of growth and adaptability, ensuring that ED teams were well-equipped to deliver high-quality care to this vulnerable patient population.

Data Measures

To assess the impact of the PPED initiative, specific outcome, structure, and process measures were established. These measures focused on identifying and managing postpartum patients at risk of preeclampsia, ensuring timely and appropriate care, and fostering collaboration between emergency and obstetric departments. Below are the key data measures tracked throughout the project. To see the preliminary results of these measures, please refer to the appendices.

Outcome Measures:

1. All patients screened as positive for postpartum status will be screened for elevated blood pressure SBP ≥ 140 and/or DBP ≥ 90. *
2. All joint ED and OB units will perform case review for all cases of postpartum patients presenting with elevated blood pressure SBP ≥ 140 and/or DBP ≥ 90 through the emergency department. *

Structure Measures:

1. All emergency departments (ED) and L&D units will have a policy outlining the triage, consultation, admission, and treatment process for postpartum patients presenting with elevated blood pressure SBP ≥ 140 and/or DBP ≥ 90 and symptoms of preeclampsia.
2. All ED physicians and Advanced Practice Providers will be educated on the hospital unit plans and protocols for postpartum patients presenting with elevated blood pressure SBP ≥ 140 and/or DBP ≥ 90 and symptoms of preeclampsia in the ED.
3. All ED nurses will be educated on the hospital unit plans and protocols for postpartum patients presenting with elevated blood pressure SBP ≥ 140 and/or DBP ≥ 90 and symptoms of preeclampsia in the ED.

Process Measures:

1. All EDs will screen all patients who can become pregnant for pregnancy/postpartum status to identify those at risk of preeclampsia. *
2. All postpartum patients with persistent severe range blood pressure SBP ≥ 160 and/or DBP ≥ 110 are treated within 60 minutes of the first elevated blood pressure. *
3. All EDs will increase the timely transfer of postpartum patients with persistent severe range blood pressures SBP ≥ 160 and/or DBP ≥ 110 as designated by their hospital policy. *

*Outcome and Process measures disaggregated by Race, Ethnicity, and Language. Measures disaggregated by geographic location (Cohort, LoMC, and Urbanization level).

Key Accomplishments

While bridging collaboration between obstetric and emergency departments, we learned some of the challenges of treating postpartum preeclampsia, which led to several important developments:

Development of Texas Maternal Centers Map

One of the obstacles rural hospitals and Level I and II maternal centers shared was that, at times, they struggled to know where to obtain an OB consult or where to transfer a patient in an emergency for higher-level care. In response to this feedback, TCHMB, in conjunction with the PPED workgroup, created the Texas Maternal Center Map (https://www.tchmb.org/pped#map), an interactive map that shows maternal hospital levels of care and their contact information across Texas. Clicking or hovering over the map will show the RAC-PCR- name, hospital name, level of care, and transfer center phone number. The hospital maternal (OB) care levels are color-coded to match the 4 different levels of care [1- Basic Care; 2- Specialty Care; Subspecialty Care; 4- Comprehensive Care).

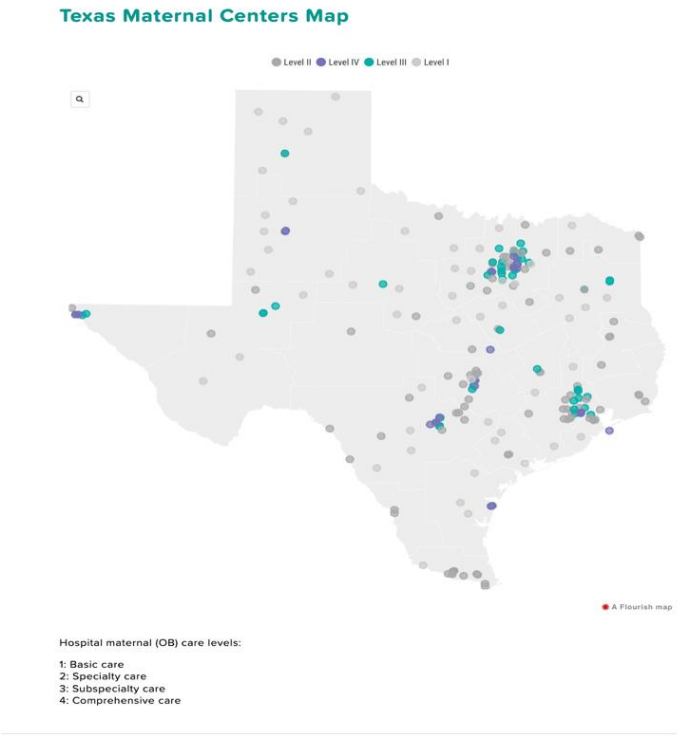

Figure 3: Texas Maternal Centers Map

Broad Education Outreach

To support hospitals in engaging their EDs in obstetric care, particularly regarding postpartum preeclampsia, TCHMB hosted a webinar titled *Preventing Maternal Mortality and Morbidity: Postpartum Preeclampsia in the ED*. This webinar, presented in partnership with the Texas College of Emergency Physicians, Texas Emergency Nurses Association, and University Health, was led by Dr. Carey Eppes and TCHMB's senior QI nurse, Susan Dimitrijevic, in December 2023 (**go.uth.edu/pped-webinar**).

Following the webinar, a recording was posted on the TCHMB website and YouTube channel, which garnered significant attention. Global Medical Response (GMR), a company that provides transport services for all types of patients, including pregnant and postpartum women, requested the webinar to be presented to their clinical staff in February 2024. This presentation reached attendees across the United States.

In addition, VuMedi, a video education platform with over 500,000 clinician members worldwide (https://www.vumedi.com/), contacted TCHMB to request permission to post the webinar on their platform, further extending its reach and impact.

Dr. Carey Eppes also presented the PPED project at the 2024 national ACOG Annual Clinical and Scientific Meeting, highlighting the successes and challenges of implementing a QI initiative that involved a department outside the traditional perinatal sphere.

Sharing of Resources Between Hospitals

The PPED initiative was successful in the sharing of resources and best practices between participating hospitals. To facilitate data collection, hospitals were grouped based on the EHR system used in their ED. This allowed them to share metrics and strategies for pulling reports, particularly in cases where postpartum screening had not yet been implemented. Additionally, hospitals shared educational materials provided to ED staff, discussed which questions to include in their EHR systems once screening was established, and collaborated on case reviews and other processes. This collective effort not only streamlined data collection but also fostered a community of learning and support among the hospitals involved.

Interpretation of Preliminary Findings

This section presents the preliminary analysis of outcome, structure, and process measures tracked throughout the project. The following graphs and figures provide a summary of the initial findings.

A total of 644 de-identified patients were submitted over the course of the project. Table 1 below displays the number of patients submitted for baseline (aggregate of previous 6 months data before initiation of PPED), then monthly throughout the project to show the monthly (n) for Figures 5, 6, 12, 13, 14 & 17):

	Base	April '23	May '23	June '23	July '23	Aug '23	Sept '23	Oct '23	Nov '23	Dec '23	Jan '24	Feb '24	Mar '24	April '24
Number of hospitals who submitted data.	20	13	13	14	11	15	14	15	8	11	9	7	8	6
Number of patients submitted.	169	41	40	26	43	50	45	41	27	43	13	22	21	20
Patient counts that had 2 *EBPs by month	60	14	10	9	15	18	16	26	11	12	4	5	9	12

Table 1: Monthly (n) for (figures 5-6, 12-14 & 17)
*Elevated Blood Pressure (EBP)

***Outcome Measure 1*: All patients screened as positive for postpartum status will be screened for elevated blood pressure (SBP ≥ 140 and/or DBP ≥ 90).**

Hospitals participating in the initiative aimed to ensure that all postpartum patients were screened for elevated blood pressure during their ED visits. However, not all hospitals had policies in place at the start of the project, and efforts were focused on addressing these gaps through the implementation of structure measures. While progress was made in putting these policies into place, consistent application of the screening processes still varied, impacting the ability to fully meet this goal. It is also important to note that this data has not yet been fully analyzed (see Figure 5). Future analyses will disaggregate outcome and process measures by race, ethnicity, and language to identify potential disparities in care. Measures will also be disaggregated by geographic location, including Cohort, Level of Maternal Care (LoMC), and urbanization level, to better understand how these factors influence screening practices and outcomes.

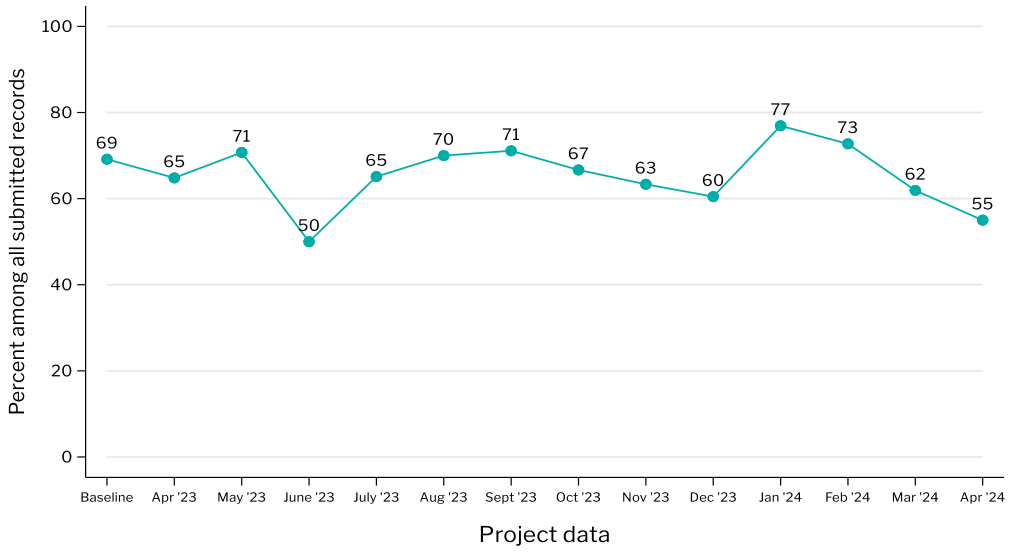

Figure 5: Percent of Submitted Patients with Repeated Blood Pressure Taken During ED Stay.

***Outcome Measure 2*: All joint ED and OB units will perform case reviews for all cases of postpartum patients presenting with elevated blood pressure (SBP ≥ 140 and/or DBP ≥ 90) through the emergency department.**

The goal of ensuring that all joint ED and OB units performed case reviews for postpartum patients with elevated blood pressure was not met. Hospitals faced challenges in establishing consistent collaboration between ED and OB departments, which impacted the ability to review every case. Strategies were implemented to foster relationships between ED and OB leadership and staff, and resources were provided to support conducting case reviews that involved multiple units. However, further work is required to improve communication and coordination to achieve the goal of reviewing all cases. It is also

important to note that this data has not yet been fully analyzed (see Figure 6). Future analyses will disaggregate data by race, ethnicity, language, and geographic location to ensure equitable care and case review practices (see Figure 6).

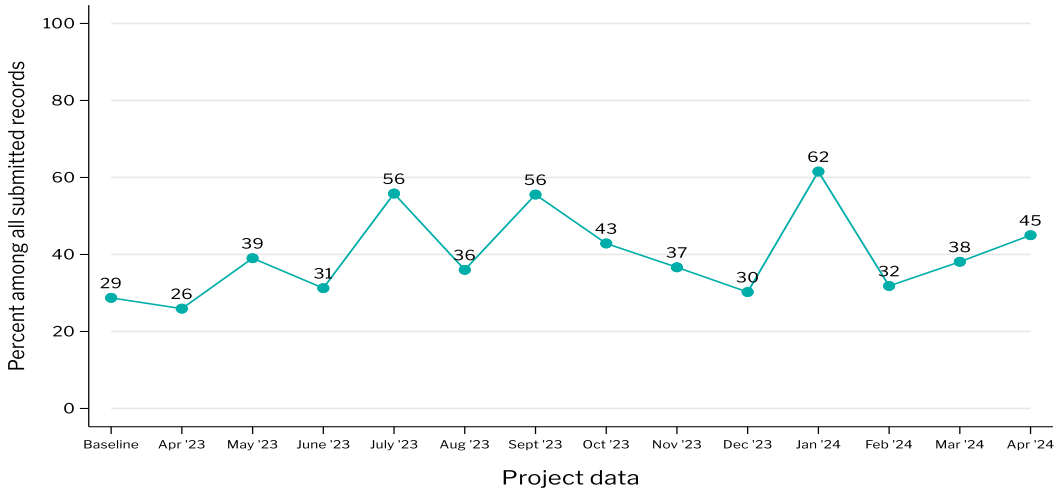

Figure 6: Percent of Patients that had a Joint ED and OB Case Review Performed

Structure Measure Assessment

Hospitals participating in the PPED initiative were tasked with implementing key structure measures related to the triage, consultation, admission, and treatment of postpartum patients presenting with elevated blood pressure. These structure measures were evaluated through surveys conducted at both the beginning (intake) and end (post-project) stages of the initiative.

The surveys provided insight into the hospitals' progress in adopting these policies and practices, enabling an assessment of how well they implemented and integrated the required processes into their routine care. Survey responses were based on a Likert scale, which provided a standardized measure of progress. A detailed description of the Likert scale and survey methodology is available in **Appendix B.**

This approach allowed for a comparison of hospital performance between the intake and post-project stages, illustrating how hospitals advanced from planning to the full implementation of these policies. The following figures represent the progress made during the initiative:

Structure Measure 1a: Progress in Implementing a Standard Process for Timely Triage and Evaluation of Pregnant and Postpartum Patients

At the start of the project, hospitals averaged a score of 3.1 on the Likert scale, indicating that triage processes for pregnant and postpartum patients were only started on a small scale. By the end of the project, this score had increased to 4.5, reflecting considerable progress toward full implementation. However, the goal of having these policies routinely used for all patients was not fully achieved. (See **Appendix B** for details on the Likert scale used).

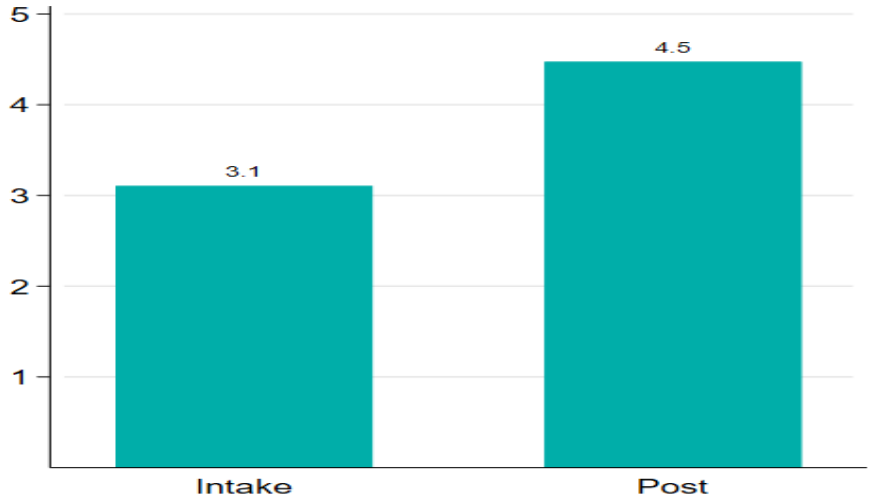

Figure 7: Standard process for timely triage and evaluation of pregnant and postpartum patients presenting with elevated blood pressure SBP ≥ 140 and/or DBP ≥ 90 and symptoms of preeclampsia. (n = 23)

Structure Measure 1b: Progress in Implementing a Policy or Guidelines for the Consultation Process of Postpartum Patients

The average score at intake for hospitals was 3.3, indicating that consultation guidelines for postpartum patients were in the early stages of implementation. By the end of the project, this score had risen to 4.3, reflecting that many hospitals had put these policies in place, although they were not yet fully integrated into routine care (see **Appendix B** for details on the Likert scale used).

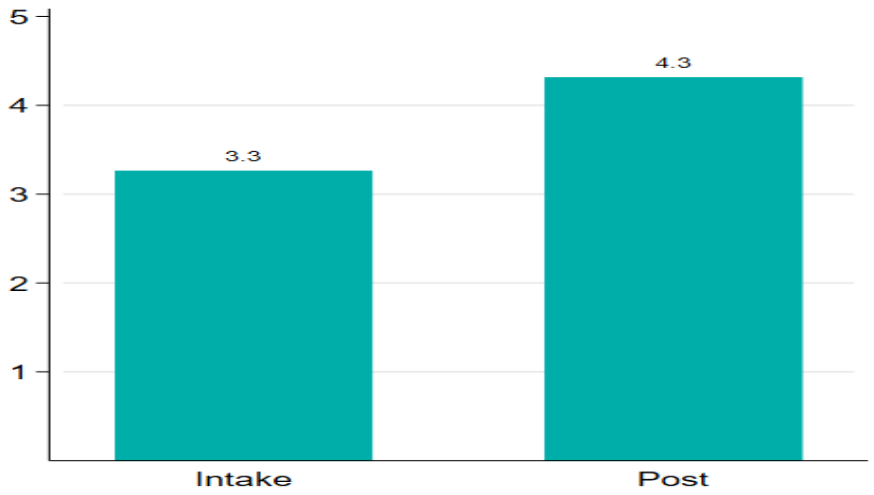

Figure 8: A policy or guidelines outlining the consultation process for postpartum patients presenting with elevated blood pressure SBP ≥ 140 and/or DBP ≥ 90 and symptoms of preeclampsia. (n = 23)

Structure Measure 1c: Progress in Implementing a Policy or Guidelines for the Admission Process of Postpartum Patients

At intake, the average score for admission policies was 3.1, indicating that only minimal implementation had taken place. By the end of the project, this score increased to 4.3, reflecting progress toward adopting these policies. However, further work is needed to ensure their routine use for all patients (see **Appendix B** for details on the Likert scale used).

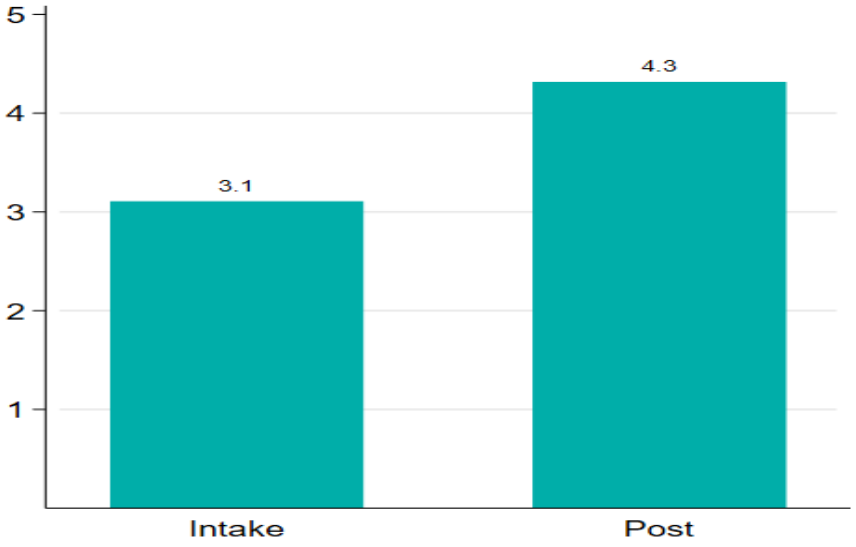

Figure 9: A policy or guidelines outlining the admission process for postpartum patients presenting with elevated blood pressure SBP ≥ 140 and/or DBP ≥ 90 and symptoms of preeclampsia (n = 23).

Structure Measure 1d: Progress in Implementing a Policy or Guidelines for the Treatment Process of Postpartum Patients

Hospitals started with an average score of 3.2, indicating that treatment guidelines were being implemented on a small scale. By the end of the project, the average score rose to 4.4, demonstrating significant progress in putting treatment guidelines into practice. However, routine application for all patients was not yet achieved. (see **Appendix B** for details on the Likert scale used).

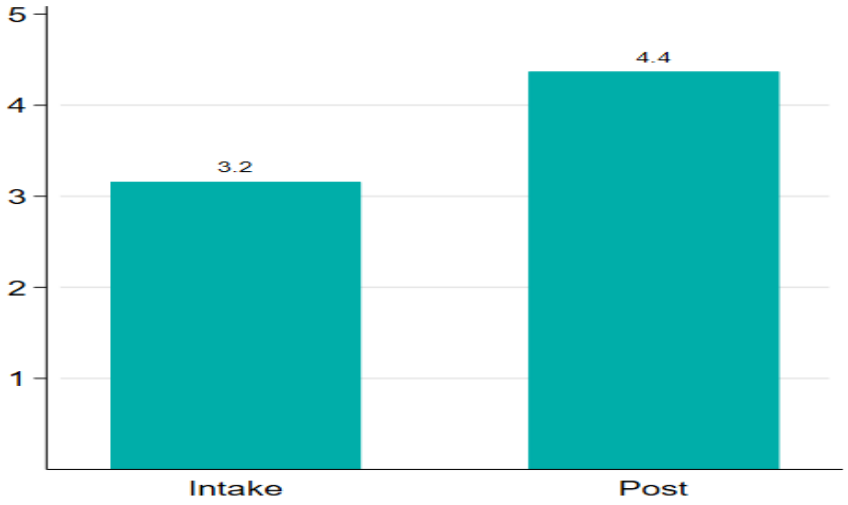

Figure 10: A policy or guidelines outlining the treatment process for postpartum patients presenting with elevated blood pressure SBP ≥ 140 and/or DBP ≥ 90 and symptoms of preeclampsia (n = 23).

Structure Measures 2 & 3: Education of ED Physicians, Advanced Practice Providers, and Nurses on Postpartum Hypertension Protocols

Hospitals aimed to educate all ED physicians, advanced practice providers, and nurses on hospital unit plans and protocols for postpartum patients presenting with elevated blood pressure (SBP ≥ 140 and/or DBP ≥ 90) and symptoms of preeclampsia in the ED.

At the beginning of the project, the intake surveys indicated that 34% of physicians and advanced practice providers and 48% of nursing staff had received education on these protocols. By the end of the project, education efforts had significantly increased, with 60% of physicians and advanced practice providers and 74% of nurses having been trained on the appropriate protocols (see Figure 11).

While considerable progress was made, the goal of ensuring that all staff members were fully educated was not completely met. Continued efforts are needed to ensure that education on these protocols is provided consistently across all ED staff.

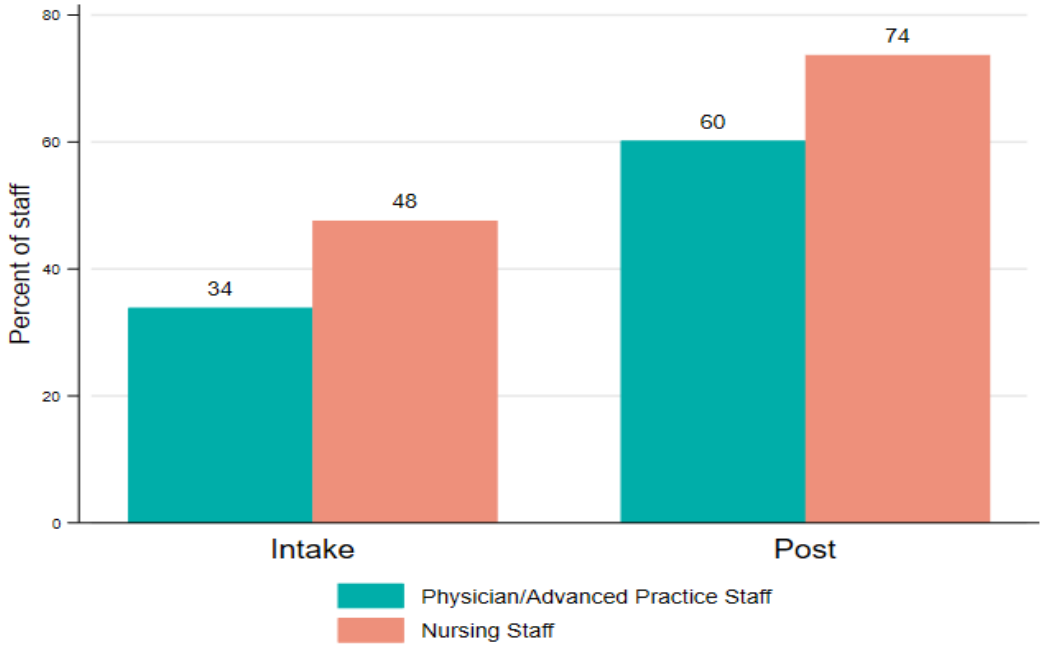

Figure 11: Percent of staff that have received education on the plans and protocols for postpartum patients with hypertension in the ED (n=23).

Process Measure 1: All EDs will screen all patients who can become pregnant for pregnancy/postpartum status to identify those at risk of preeclampsia.

The goal of this measure was to ensure that all women of childbearing age presenting to the ED were screened for postpartum status to identify those at risk of preeclampsia. However, due to limitations in the ability of many hospitals to collect the necessary data from their ED systems, the measurement strategy had to be adjusted. Instead, the project pivoted to tracking patients who were screened upon arrival and were later identified as postpartum with symptoms of preeclampsia.

To support this effort, hospitals worked to integrate screening tools into their Electronic Health Records (EHRs) to standardize the process for identifying postpartum status in women of childbearing age. These efforts are ongoing, and many hospitals continue to collaborate with their EHR teams to fully implement these tools. Additionally, education was provided to ED staff and providers to emphasize the importance of understanding postpartum status and incorporating postpartum complications into differential diagnoses.

It is important to note that this data has not yet been fully analyzed (see Figure 12). Future analyses will disaggregate outcome and process measures by race, ethnicity, and language to identify potential disparities in care. Measures will also be disaggregated by geographic location, including Cohort, Level of Maternal Care (LoMC), and urbanization level, to better understand how these factors influence screening practices and outcomes.

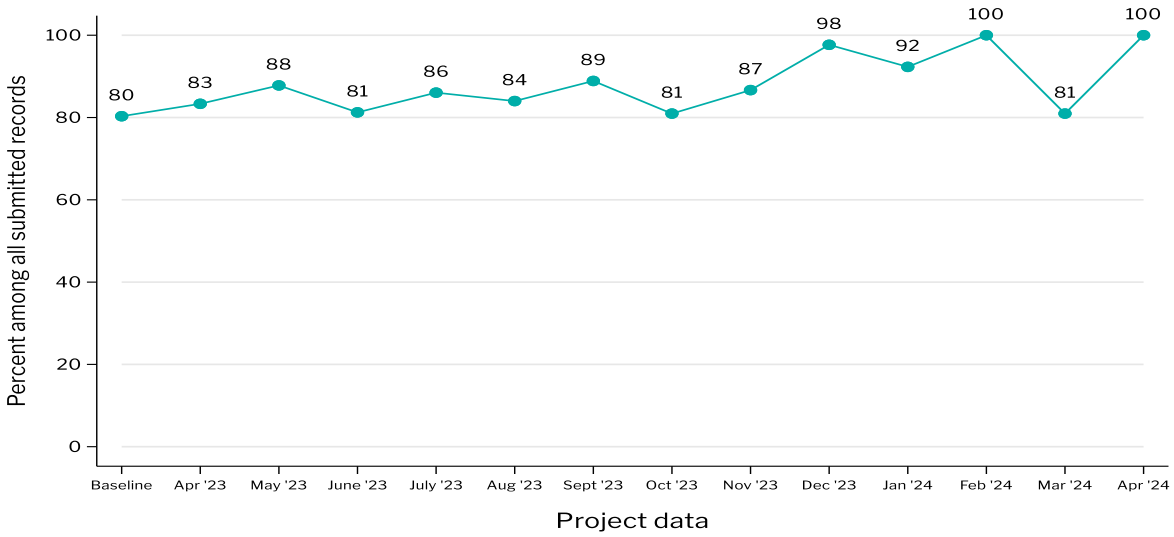

Figure 12: Percent of Submitted Patients Screened for Postpartum Status on ED Arrival.

Process Measure 2: All postpartum patients with persistent severe range blood pressure (SBP ≥ 160 and/or DBP ≥ 110) are treated within 60 minutes of the first elevated blood pressure.

The goal of this measure was to ensure that all postpartum patients with severe range blood pressure were treated within 60 minutes of the first elevated reading. Hospitals worked to improve timely intervention by ensuring quick response and treatment protocols were in place for these patients.

Efforts were made to implement standardized treatment guidelines, and hospitals educated ED staff on the importance of prompt treatment to prevent complications from severe hypertension in postpartum patients. The data shows the progress hospitals made over the course of the project (see Figure 13).

As with other measures, this data has not yet been fully analyzed. Future analyses will disaggregate the results by race, ethnicity, and language to identify any potential disparities in treatment. Additionally, measures will be broken down by geographic location, including Cohort, Level of Maternal Care (LoMC), and urbanization level, to better understand how these factors may influence the timely treatment of patients with severe hypertension.

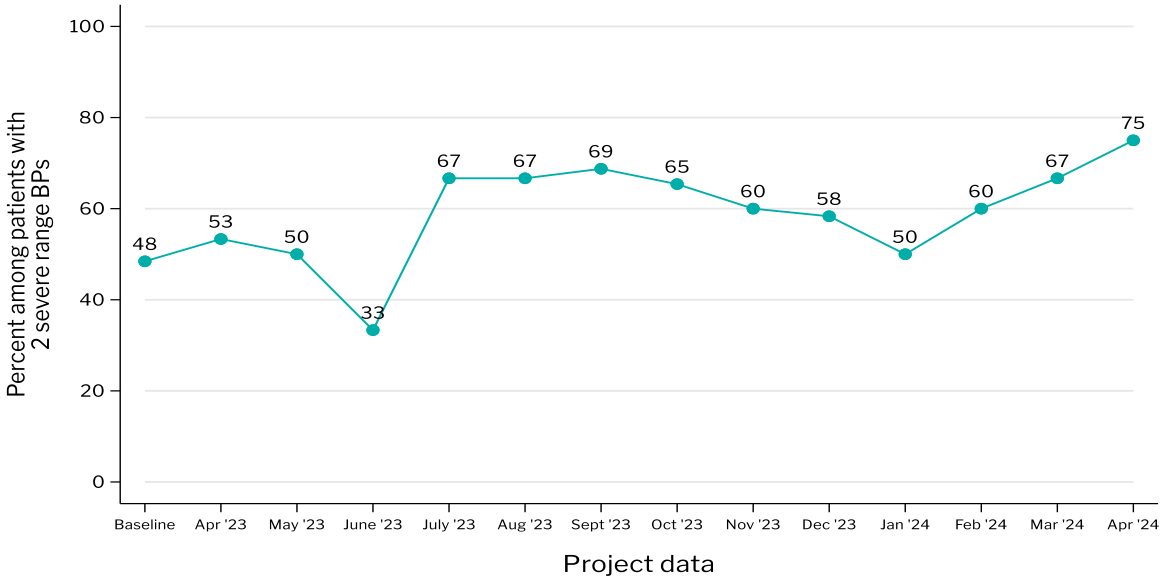

Figure 13: Percent of Patients with 2 Severe Range BPs Treated with an Antihypertensive within 1 Hour of 1st Severe Range BP.

Process Measure 3: All EDs will increase the timely transfer of postpartum patients with persistent severe range blood pressures (SBP ≥ 160 and/or DBP ≥ 110) as designated by their hospital policy.

The goal of this measure was to ensure that all postpartum patients with persistent severe range blood pressures were transferred in a timely manner as designated by hospital policy. Hospitals worked to improve the timely transfer of these patients by enhancing coordination between the ED and OB teams and ensuring adherence to hospital protocols for patient transfer.

The results are shown in the following figures:

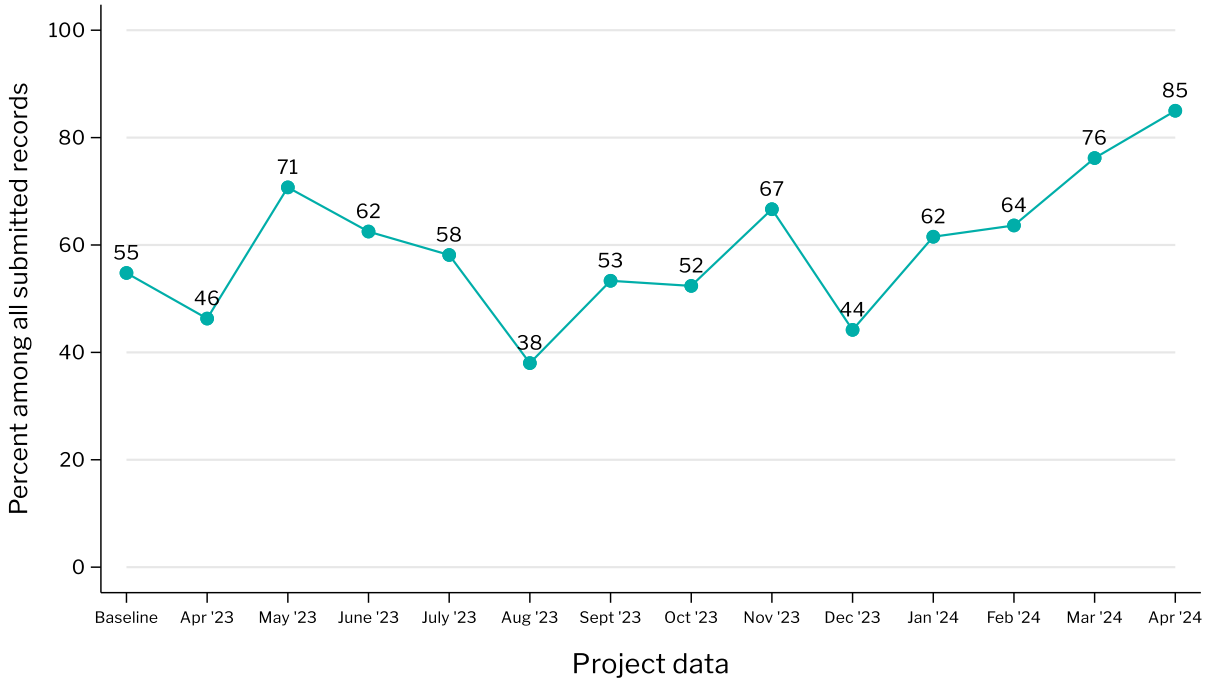

Figure 14: Percent of Submitted Patients who had an OB Consult

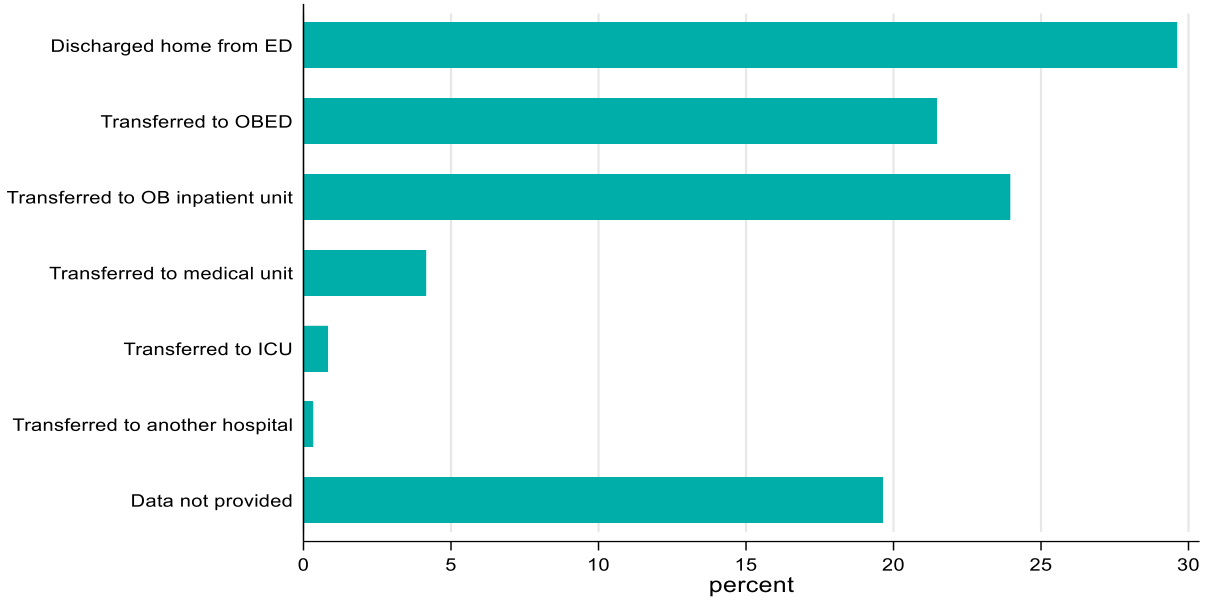

Figure 15: Disposition* of Patients from the Emergency Department

*(*Where patient was sent when leaving the Emergency Department*).

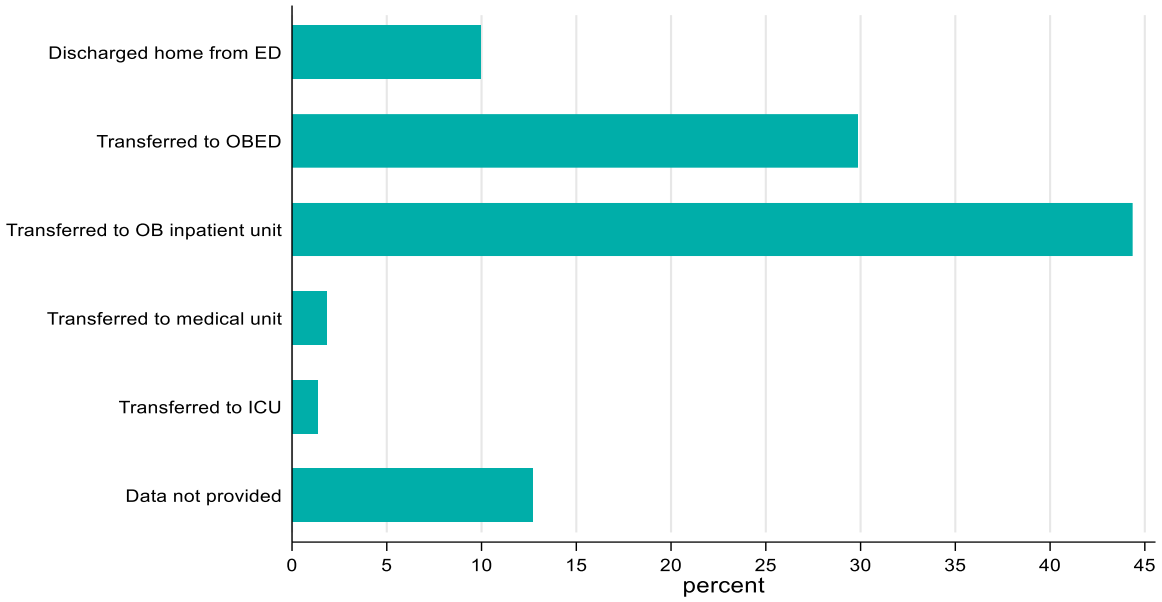

Figure 16: Disposition* of Patients from the Emergency Department among those with 2 severe range BPs. **(Where patient was sent when leaving the Emergency Department).*

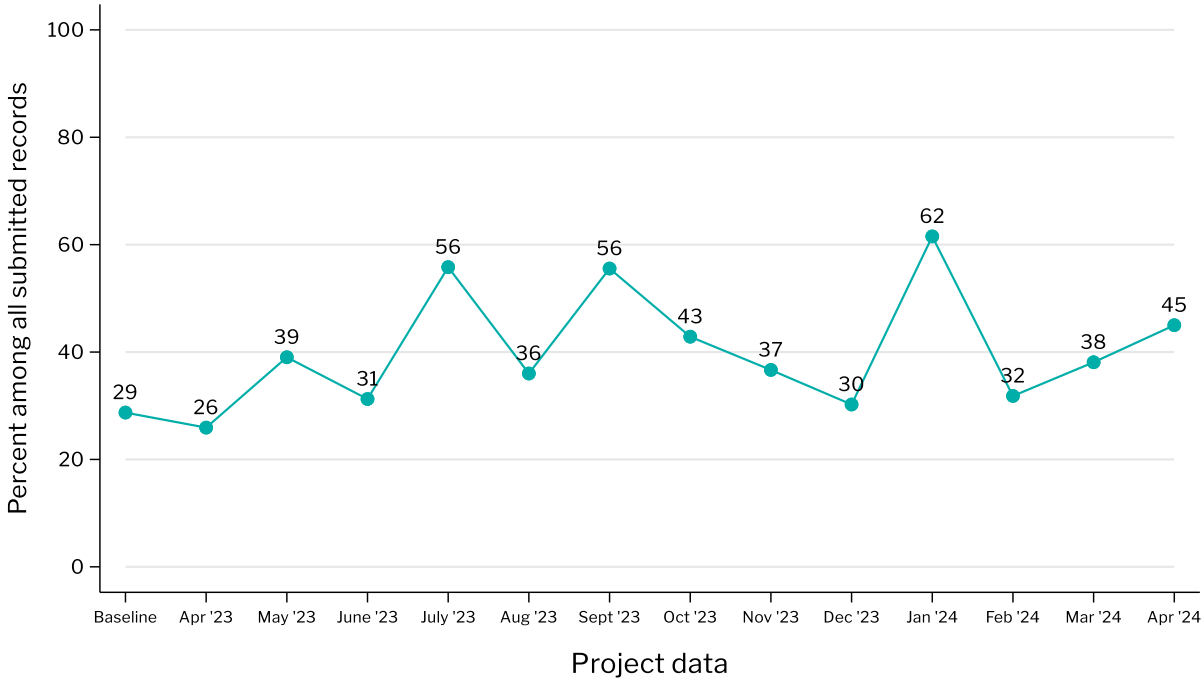

Figure 17: Percent of Patients with 2 Severe Range BPs that had an OB consult within 1 Hour of 1st Severe Range BP.

Hospitals made efforts to streamline the transfer process, educating ED staff on the importance of OB consults and strengthening communication between departments. These measures were in place to help improve the overall response to patients presenting with severe range blood pressure.

It is also important to note that the data presented here has not yet been fully analyzed. Future analyses will disaggregate the results by race, ethnicity, language, and geographic location (including Cohort, Level of Maternal Care (LoMC), and urbanization level) to ensure equitable care and better understand the factors influencing timely transfers.

Conclusion and Future Directions

The PPED initiative has uncovered fundamental gaps in the hospital processes used to identify, treat, and manage postpartum preeclampsia. These gaps highlight critical areas where further QI efforts are needed. A significant challenge identified through this project is the lack of routine screening for postpartum status in EDs, which obscures the true magnitude of postpartum women seeking care for related complications. This gap emphasizes the need for continued focus on enhancing/standardizing screening processes to better understand and address the needs of postpartum women in ED settings.

While we have concluded the data collection phase of the PPED initiative, TCHMB remains committed to completing the final phases of this critical project, as referenced in the PPED Dissemination Plan. Our immediate focus is on analyzing the collected data to provide comprehensive reports at the hospital, RAC, and state levels. Future efforts will also involve detailed analyses to disaggregate outcome, structure, and process measures to identify any disparities and uncover further opportunities for improvement. The findings from these analyses, along with key insights from the PPED project, will be disseminated widely to inform best practices and support other hospitals in improving maternal and postpartum care. Information gleaned through the PPED project will be developed into a toolkit, including best practices, lessons learned, and resources created during the project.

Over the next year, we will complete the final phase of the project. This includes data analysis, report generation, toolkit creation, and publication. Our top priority is ensuring that all hospitals, regardless of size or location, have access to the vital tools needed to enhance the quality of care for postpartum patients. Furthermore, TCHMB plans to release a comprehensive PPED toolkit to be made available to stakeholders on how to build and foster interdepartmental relationships between the OB and ED departments to improve postpartum preeclampsia maternal care.

Appendix

Appendix A: ***Example of resource to foster OB and ED relationship***

Recommendations for ED and OB Partnership

- **Step 1: Create a relationship;**

- **Step 2: Recognize all the specialties needing EDs attention;**

- **Step 3: Collaborate: Help us Help you!**
  - **Recognition of Maternal HTN, but also stroke triggers for OB patients.**
    - **What is normal for other ED patients is high risk for OB.**
    - **Meet in the middle for what and who is best for each problem to address; and**

- **Step 4: Stay Humble. Stay Fair. It's a Process**
  - **Rough Patches – but working it out as they come.**
  - **Transparent and have patience.**
  - **Keep eyes to quality and patient experience and safety.**
  - **Celebrate WINS.**

Recommendations provided by the Titus Regional Hospital OB and ED department staff at the April PPED Reinforcement call.

Appendix B: Likert Scale Description and Survey Methodology

As part of the PPED initiative, hospitals completed surveys at both the intake and post-project stages to assess their progress in implementing policies related to the project. The surveys utilized a Likert scale with the following response options:

1. Not Yet
2. Started planning, but not yet in place;
3. Started on a small scale, but not yet in place;
4. In place, but not routinely used for all patients; and
5. Fully in place and routinely used for all patients.

This scale allowed for the evaluation of progress across various structure measures, enabling a comparison between the initiative's initial and final stages.

References

1. https://www.acog.org/womens-health/faqs/preeclampsia-and-high-blood-pressure-during-pregnancy
2. Texas Maternal Mortality and Morbidity Review Committee and Department of State Health Services Joint Biennial Report, December 2020
3. Texas Maternal Mortality and Morbidity Review Committee and Department of State Health Services Joint Biennial Report 2022
4. https://www.dshs.texas.gov/maternal-child-health/texas-data/pregnancy-risk-assessment-monitoring